DP Biology – B3.3 (HL only)

Muscle and movement

- How do muscles contract and produce movement?
- What do animals gain from having muscle tissue?

HL statements B3.3.1–10. Textbook pp. 326–340.



Image: a humpback whale breaching. Giles Laurent, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.3.1, B3.3.9	Compare motile and sessile organisms, and give reasons why animals move from place to place
2	B3.3.2	Explain how actin and myosin slide to shorten a muscle
3	B3.3.3, B3.3.8	Explain how titin and antagonistic muscles return a muscle to its length, using the intercostal muscles
4	B3.3.4	Describe a motor unit and how it controls a muscle
5	B3.3.5	Explain how skeletons anchor muscles and act as levers
6	B3.3.6–7	Explain movement at the hip joint and measure the range of motion of joints
7	B3.3.10	Explain how marine mammals are adapted for swimming

Everything moves

B3.3.1



Barnacles: sessile adults. Chris Light, CC BY-SA 4.0.

	Motile	Sessile
Meaning	Moves from place to place	Fixed in one place
Example	Humpback whale	Barnacle, coral
Movement	Whole body: swimming, walking, flying	Parts only: feeding cirri, tentacles

Key idea

Movement is a feature of all living things. Sessile animals still move parts of their bodies to feed, and often have a motile larval stage that finds a new place to settle. Even single cells move: bacteria swim with flagella, and white blood cells crawl towards an infection.

Why animals move

B3.3.9

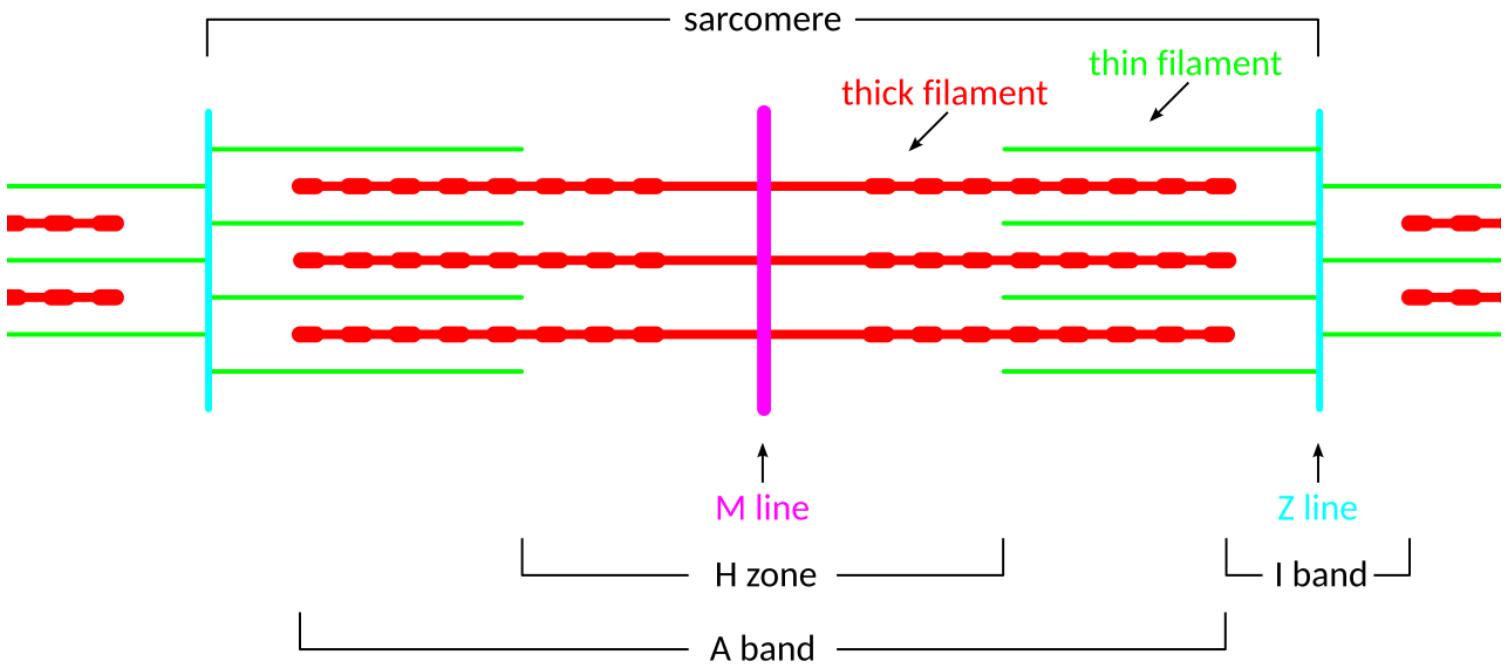
Reason	Example
Foraging for food	Honey bees fly kilometres to collect nectar and pollen
Escaping danger	Impala leap away from a lion; squid jet away using their mantle
Searching for a mate	Male moths fly towards the scent of a female
Migration	Humpback whales swim between cold feeding waters and warm breeding waters each year

Research

Find one more example of each reason, from a different animal group.

The sarcomere

B3.3.2



A sarcomere: thick (myosin) and thin (actin) filaments. CC BY-SA 3.0.

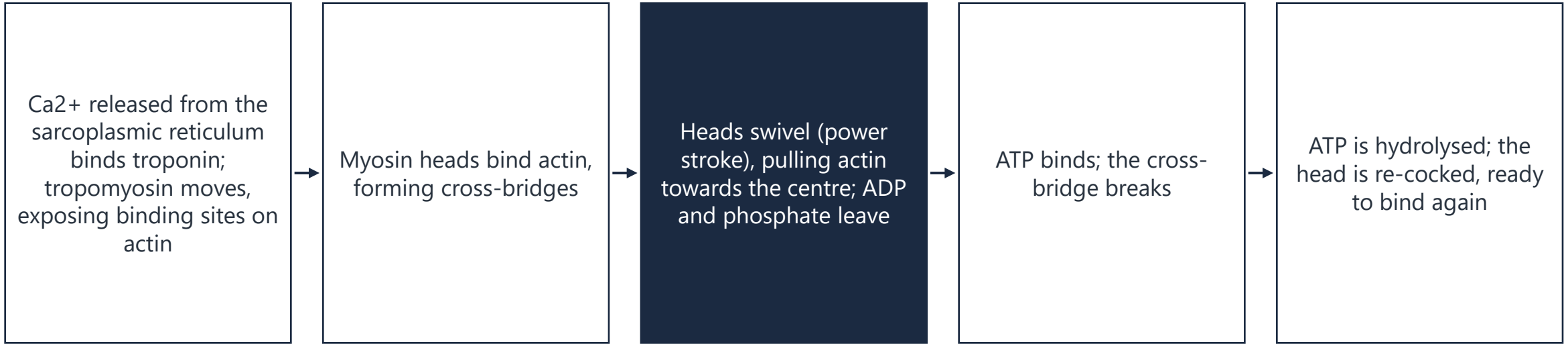
- A muscle fibre contains many myofibrils
- Each myofibril is a chain of **sarcomeres**, from one Z line to the next
- Thick filaments are **myosin**; thin filaments are **actin**
- The overlap makes the light and dark bands (striations)

In contraction

Actin filaments slide over myosin towards the centre of the sarcomere. The Z lines move closer and the light band and H zone get shorter, but the filaments themselves do not change length.

How the filaments slide

B3.3.2

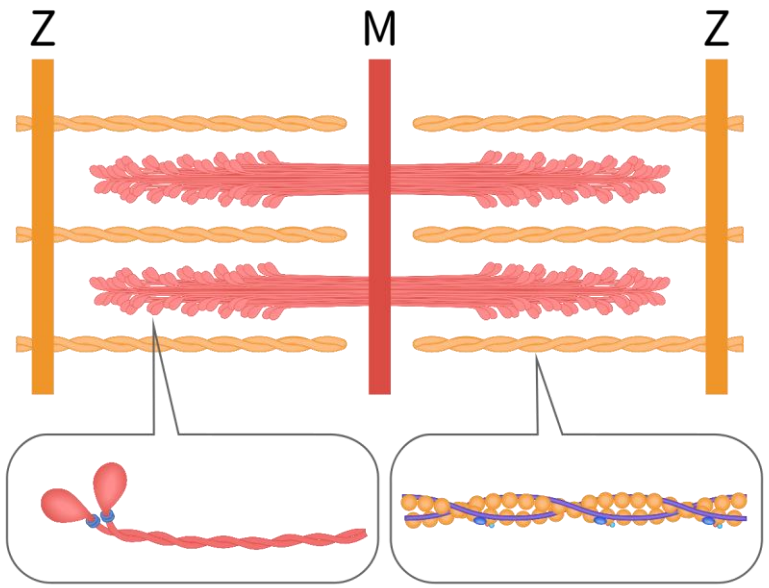


Think about it

After death, ATP runs out and muscles lock stiff (rigor mortis). Use the cycle to explain why. Then explain why contraction stops when calcium ions are pumped back into the sarcoplasmic reticulum.

Titin and antagonistic muscles

B3.3.3



- Muscles can only pull: they exert force only when they contract
- So muscles work in **antagonistic pairs**: when one contracts, it stretches the other
- **Titin**, the largest known protein, runs from the Z line to the middle of the sarcomere
- When stretched, titin stores energy like a spring and helps the sarcomere recoil
- It also stops the sarcomere being overstretched

Sarcomere detail. DBCLS, CC BY 4.0.

Example

Biceps and triceps: the biceps bends (flexes) the elbow, the triceps straightens (extends) it.

Intercostal muscles: an antagonistic pair

B3.3.8

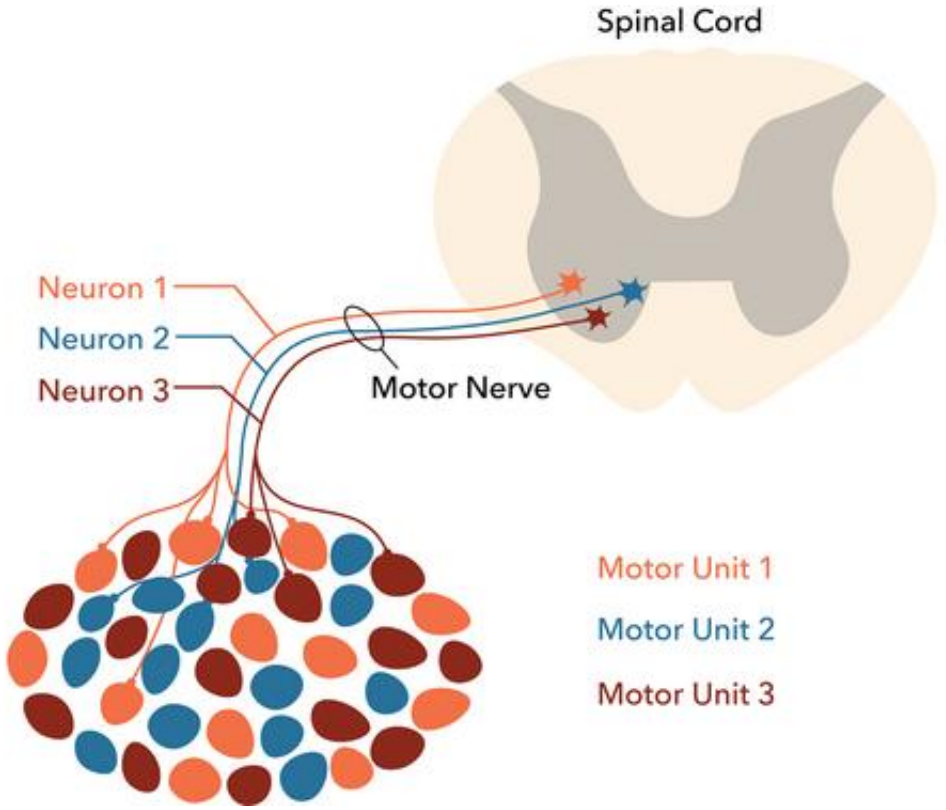
	External intercostals	Internal intercostals
Fibre direction	Run one way between ribs	Run at right angles to the external layer
When they contract	Ribs move up and out: inhalation	Ribs move down and in: forced exhalation
Effect on the other layer	Stretch the internal layer, storing energy in its titin	Stretch the external layer, storing energy in its titin

Key idea

Because the fibres in the two layers run in different directions, the same contraction moves the ribs in opposite directions. Each layer resets the other.

Motor units

B3.3.4



Three motor units in one muscle. D. Walsh and A. Sved, CC BY-SA 4.0.

- A **motor unit** is one motor neuron and all the muscle fibres it controls
- The neuron branches; each branch ends at a **neuromuscular junction** on one fibre
- Acetylcholine released there triggers contraction of the fibre
- All fibres in a unit contract together

Fine and strong control

Eye muscles have small units (a few fibres each) for precise control. Leg muscles have large units. More force comes from using more units at once.

Skeletons: anchors and levers

B3.3.5

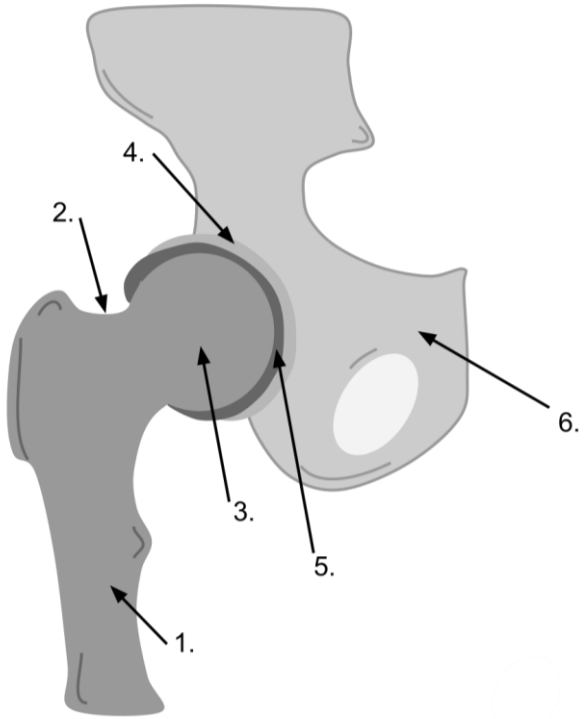
	Exoskeleton	Endoskeleton
Where	Outside the body	Inside the body
Example	Arthropods such as insects and crabs	Vertebrates
Muscles attach	To the inside of the skeleton	To the outside of bones, by tendons

Levers

A bone is a lever and the joint is the pivot (fulcrum). A muscle applies the effort where its tendon attaches, and the load is the weight being moved. Levers can change the size, speed and direction of the movement. Muscles attached close to a joint give fast, wide movements.

The hip: a synovial joint

B3.3.6



Hip joint: femur head in the pelvis. Kcotton15, CC BY-SA 4.0.

Part	Role
Femur and pelvis	The bones that move; ball (femur head) fits the socket
Cartilage	Smooth layer that reduces friction and absorbs shock
Synovial fluid	Lubricates the joint
Ligaments	Join bone to bone; stop unwanted movement
Muscles and tendons	Muscles pull; tendons join them to bones

Range of motion

Joint	Type	Movement
Hip, shoulder	Ball and socket	In three planes, and rotation
Elbow, knee	Hinge	Mainly in one plane: flexion and extension

Practical

Use a goniometer (a protractor with two arms) or an app on a phone to measure the angle through which a joint moves. Compare hip and knee, or compare people. Keep the starting position and the method the same each time.

Variables

Range of motion depends on joint type, and can be affected by age, training, injury and warming up.

Built for swimming: marine mammals

B3.3.10



Humpback whale. Giles Laurent, CC BY-SA 4.0.

Adaptation	How it helps
Streamlined body	Less drag in water
Flippers	Forelimbs for steering and balance
Fluke	Tail moved up and down for thrust (fish tails move side to side)
Blowhole	Breathe at the surface without lifting the head; airway is separate from the gut

Link

Whales and dolphins descend from land mammals. Their flippers still contain the same bones as a human arm, and the up-and-down tail movement comes from the way land mammals bend their backs when running.

Check that you can explain

1. Motile and sessile organisms, and four reasons for locomotion
2. The structure of a sarcomere and the cross-bridge cycle
3. Why muscles need antagonists, and the role of titin
4. How the intercostal muscles move the ribcage
5. What a motor unit is
6. Exoskeletons, endoskeletons and levers
7. The parts of the hip joint and the range of motion of joints
8. Adaptations of marine mammals for swimming

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the role of calcium ions in muscle contraction.
2. Explain why muscles work in antagonistic pairs.
3. State two adaptations of whales for swimming.

Image credits

001 Humpback whale breaching in Ballena Marine National Park Photo by Giles Laurent.jpg – Giles Laurent – CC BY-SA 4.0

Barnacles 2165.jpg – Chris Light – CC BY-SA 4.0

Sarcomere diagram.svg – SlothMcCarty – CC BY-SA 3.0

202107 Structure of skeletal muscle.svg – DataBase Center for Life Science (DBCLS) – CC BY 4.0

Motor unit.png – Daniel Walsh and Alan Sved – CC BY-SA 4.0

Ball and Socket Joint (Hip joint).svg – Kcotton15 – CC BY-SA 4.0

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